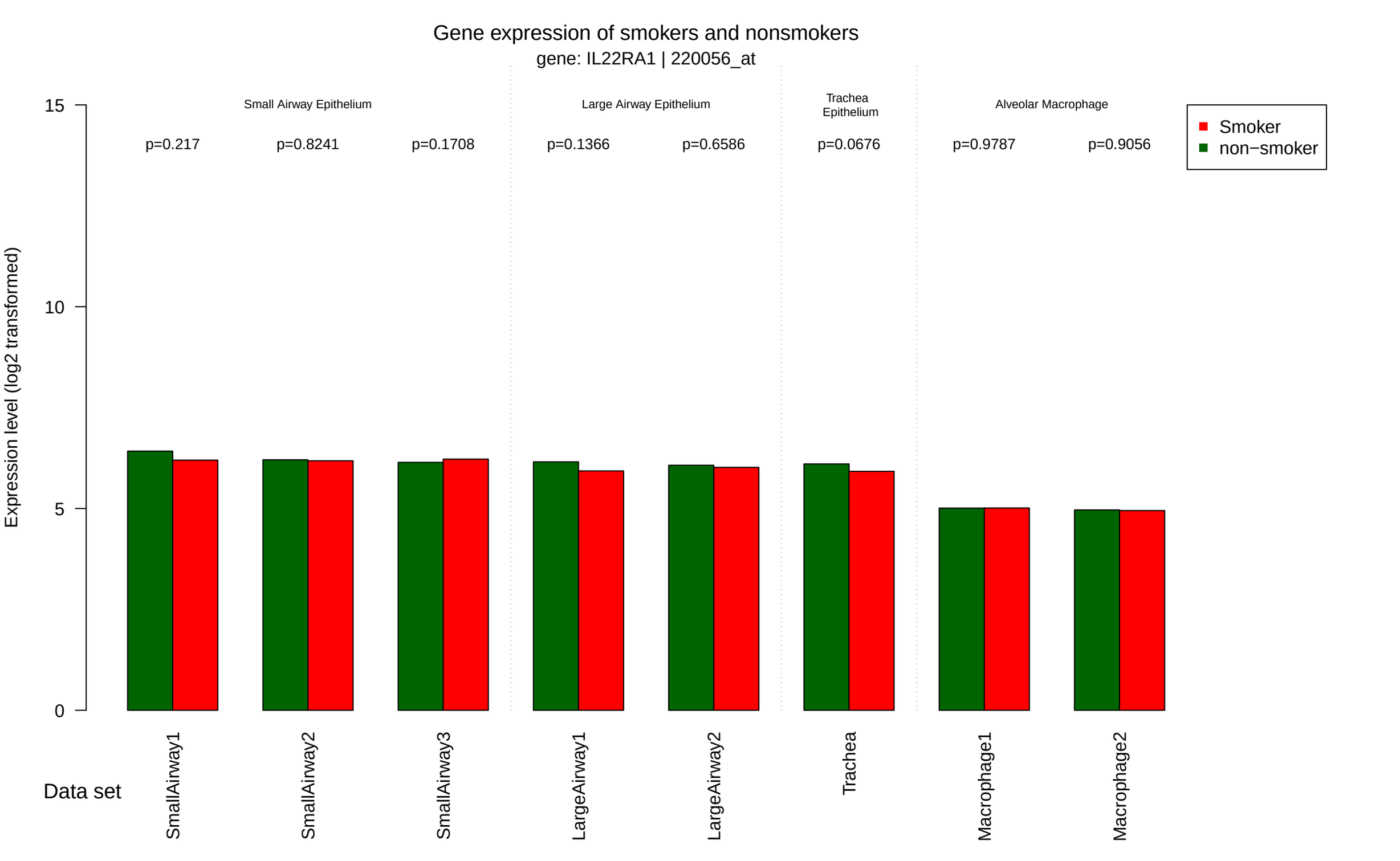
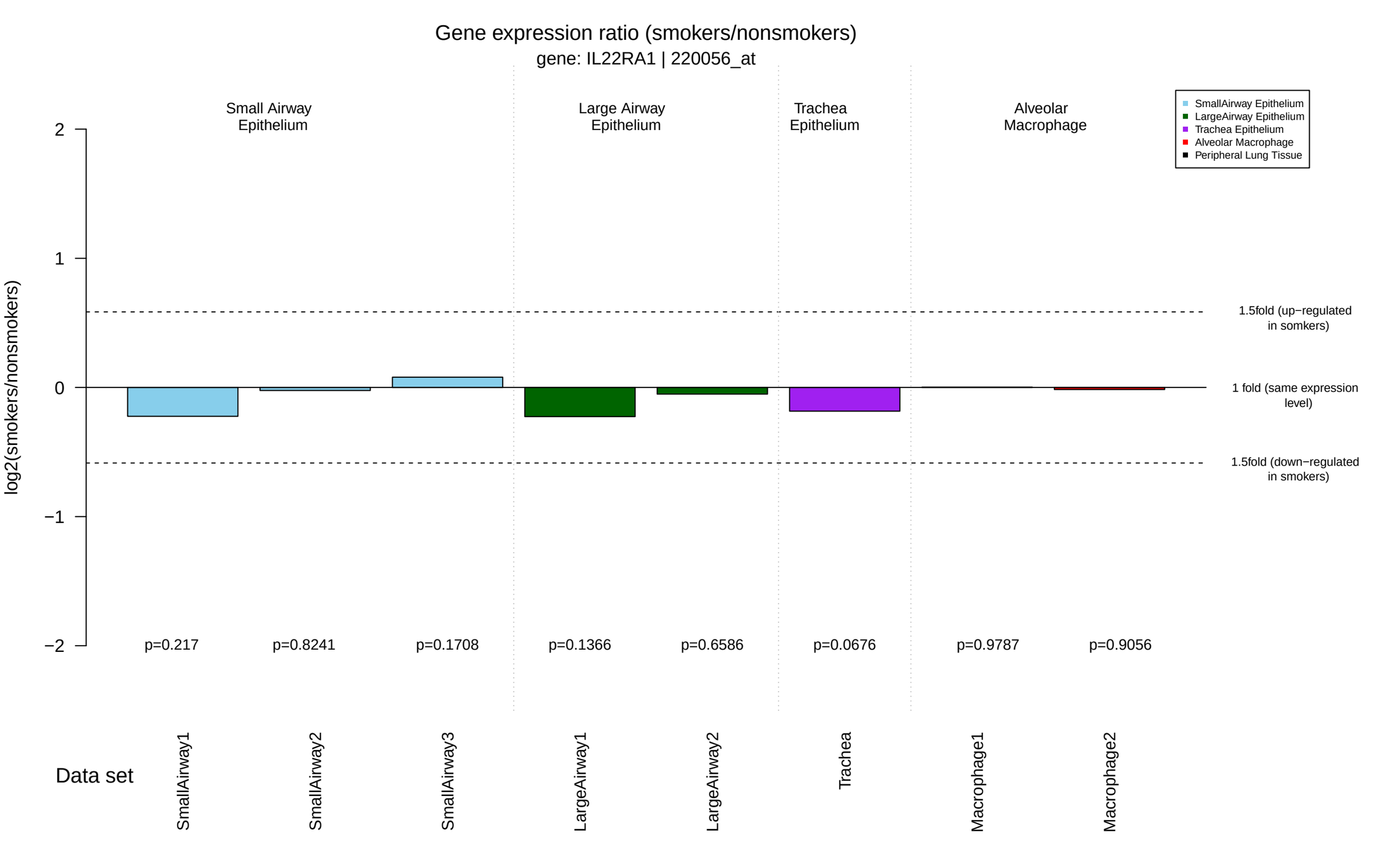
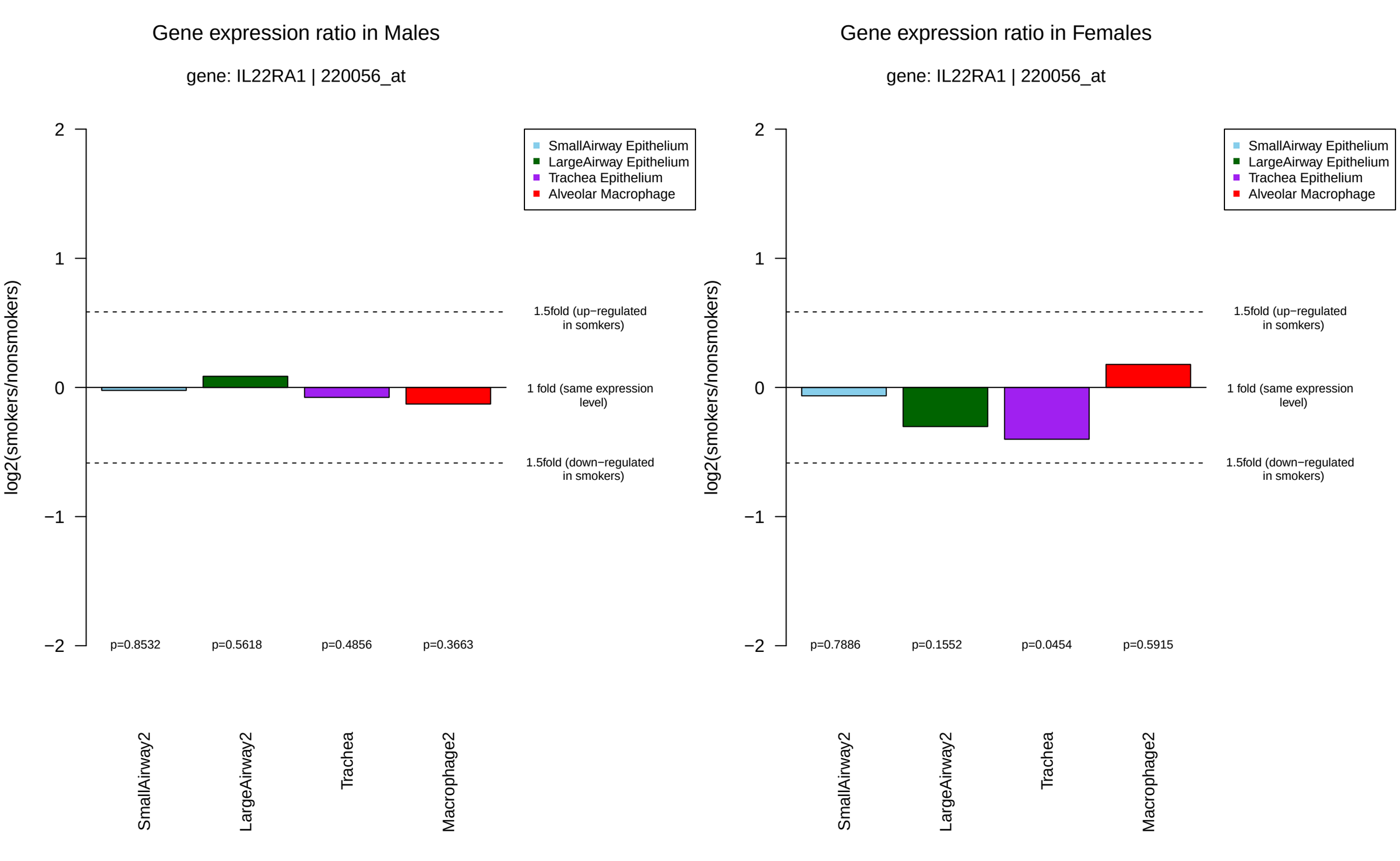
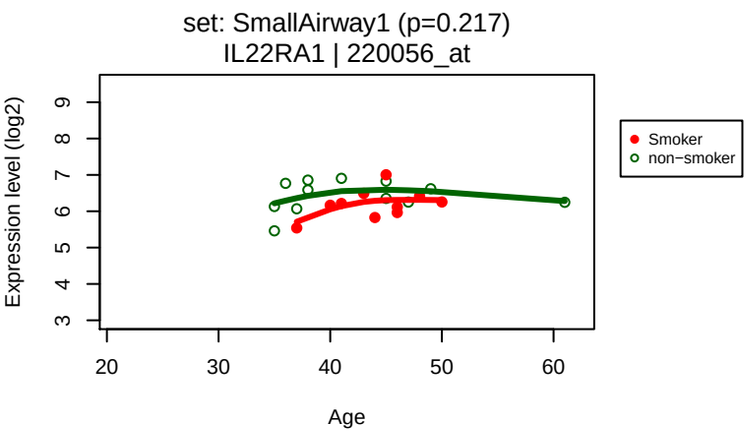
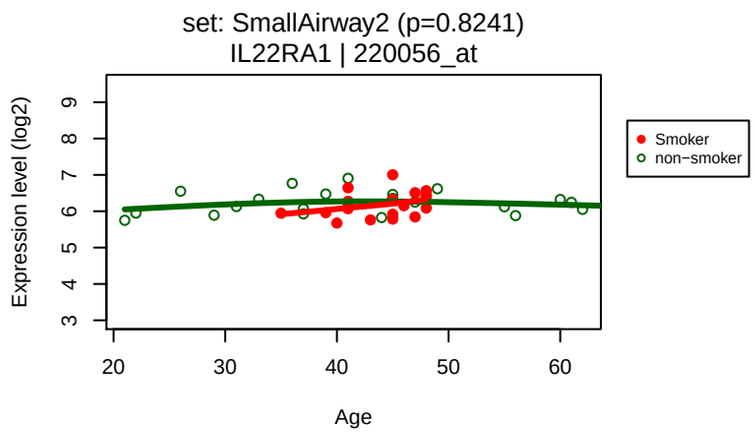
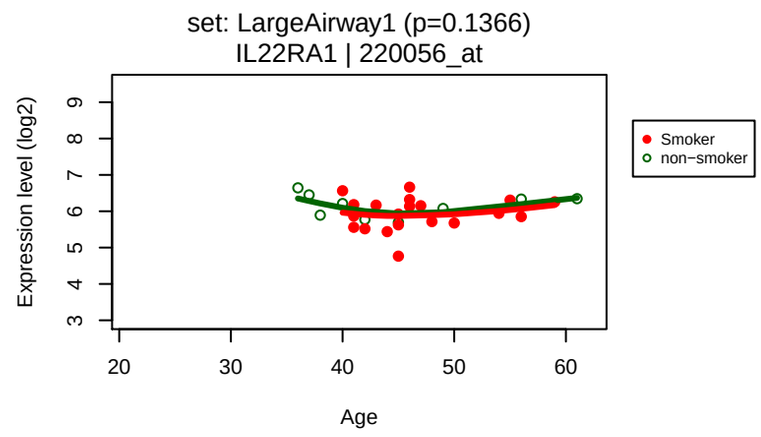

Gene expression of smokers and nonsmokers
gene: IL22RA1 | 220056_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.217
p=0.8241
p=0.1708
p=0.1366
p=0.6586
p=0.0676
p=0.9787
p=0.9056
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

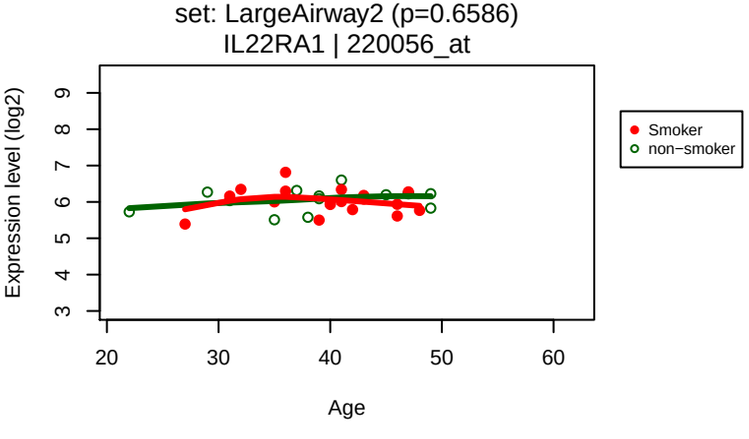

Gene expression ratio (smokers/nonsmokers)
gene: IL22RA1 | 220056_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.217
p=0.8241
p=0.1708
p=0.1366
p=0.6586
p=0.0676
p=0.9787
p=0.9056
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

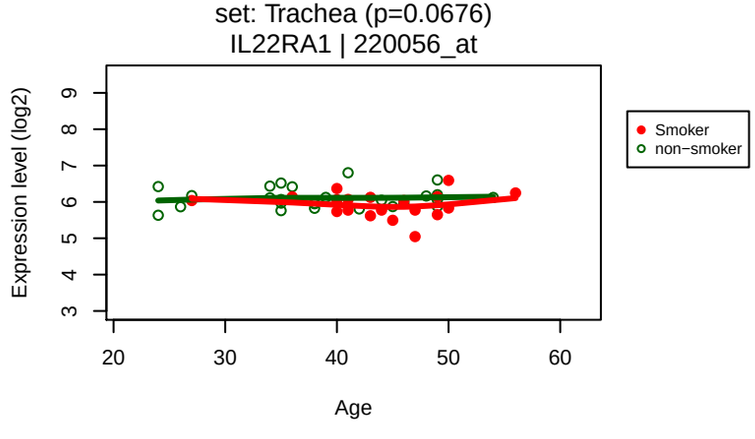

Gene expression ratio in Males
gene: IL22RA1 | 220056_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8532
p=0.5618
p=0.4856
p=0.3663
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL22RA1 | 220056_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7886
p=0.1552
p=0.0454
p=0.5915
SmallAirway2
LargeAirway2
Trachea
Macrophage2

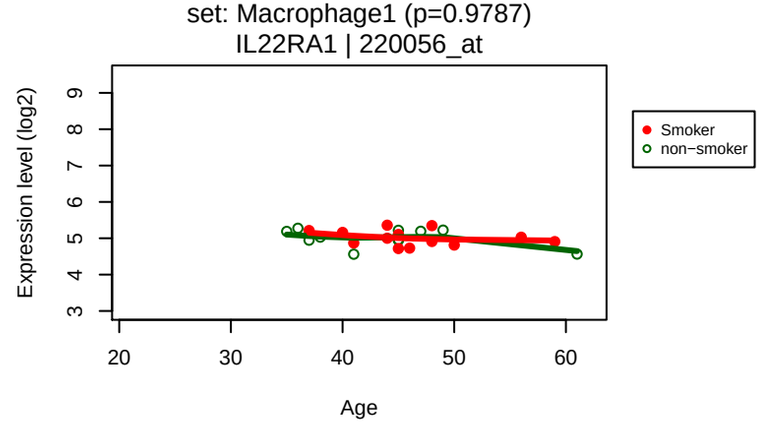

set: SmallAirway1 (p=0.217)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker

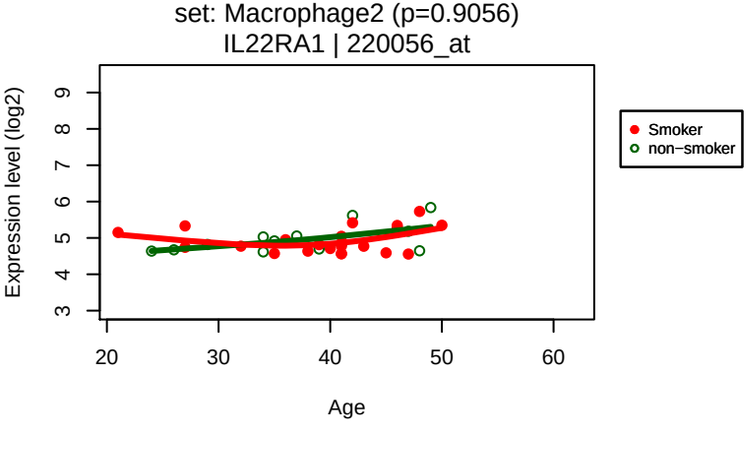

set: SmallAirway2 (p=0.8241)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker

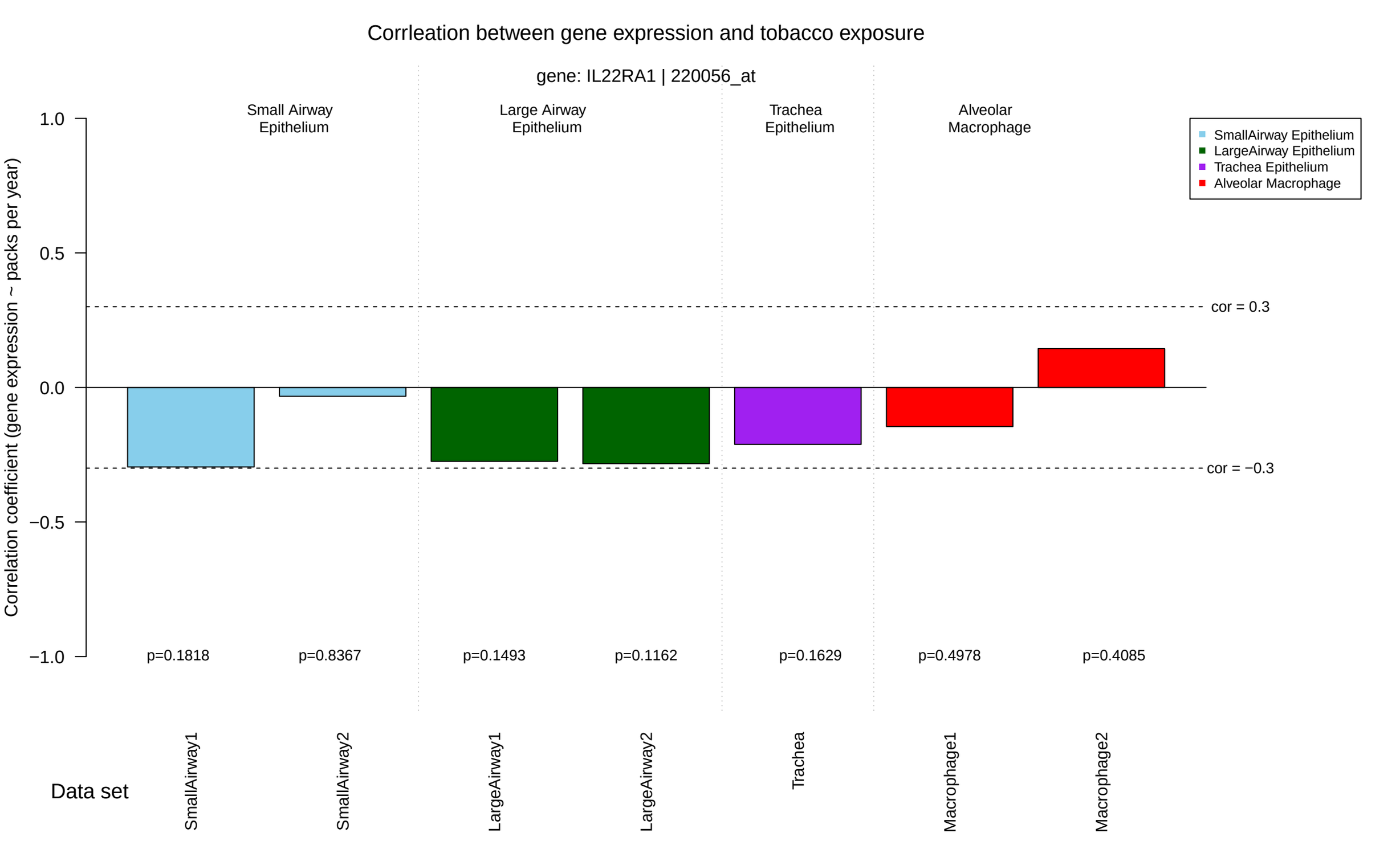

set: LargeAirway1 (p=0.1366)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6586)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0676)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9787)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9056)
IL22RA1 | 220056_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL22RA1 | 220056_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1818
p=0.8367
p=0.1493
p=0.1162
p=0.1629
p=0.4978
p=0.4085
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2